



COLLEGE OF ENGINEERING, KARUNAGAPPALLY

Department of Computer Science and Engineering

Integration and Innovation of Blockchain in Web3.0: Current Status and Standardization Prospects

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Register No.

KNP23CS094

Batch

2023–2027

Semester / Class

S7 CSE

Department

Computer Science and Engineering

SECTION 1

Introduction to Blockchain & Web3.0

Blockchain in Web3.0: Integration, Innovation & Standardization

What is Blockchain?

The foundation of trusted decentralized systems

Core Definition:

- Distributed ledger using cryptographic techniques
- Blocks linked by hash pointers — immutable chain
- No central authority; collectively managed by nodes
- Full traceability and transparency of all records

Key Properties:

- **Decentralization** — no single point of control
- **Immutability** — data cannot be altered
- **Traceability** — full audit trail



Each block contains hash of the previous block

Consensus & Security: Cryptographic hashing and distributed consensus ensure blocks cannot be altered once appended to the ledger.

Blockchain — A distributed ledger formed by appending cryptographically chained blocks confirmed by consensus — GB/T 42752-2023

What is Web3.0?

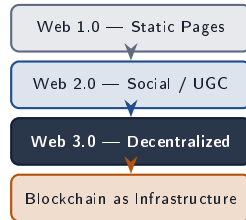
The next generation Internet built on decentralized infrastructure

Evolution of the Web:

- **Web 1.0** — Read-only; static pages, one-way info
- **Web 2.0** — Read-Write; social platforms, UGC
- **Web 3.0** — Read-Write-**Own**; decentralized, user-controlled

Web3.0 Core Features:

- Trusted collaboration via distributed ledger
- Distributed execution & smart contracts
- Data privacy & asset ownership by users
- Integrates AI, IoT, Big Data, Metaverse



Web3.0 — A blockchain-based new-generation Internet; user-owned, trustless, disintermediated; integrates information flow, business flow, and value flow

Blockchain as Backbone of Web3.0

Three core properties enabling the decentralized Internet

Decentralization

P2P Network
No central authority

- No single owner
- Nodes validate jointly
- Platform monopoly removed

Immutability

Hash-Chained Blocks
Tamper-proof storage

- Data integrity assured
- Audit trails permanent
- Fraud prevention

Traceability

Public History
Full auditability

- Every transaction visible
- Cross-border trust
- Smart contracts enabled

In Web3.0: Blockchain replaces centralized services with on-chain smart contracts, integrating information flow, business flow, and **value flow** in a single trustworthy layer.

SECTION 2

Blockchain & Big Data

Blockchain in Web3.0: Integration, Innovation & Standardization

Overview: Big Data in the Web3.0 Era

Massive, diverse, fast-moving data driving the digital economy

What is Big Data?

- Large volume, diverse types, high-speed generation
- **Structured** (relational DB), **Semi-structured** (XML/JSON), **Unstructured** (images, video)
- Processed with Hadoop, Spark; analyzed via ML

Big Data Pain Points:

- ✗ Data privacy & security risks
- ✗ Centralized storage = single point of failure
- ✗ Data ownership disputes
- ✗ Trust issues in sharing economy

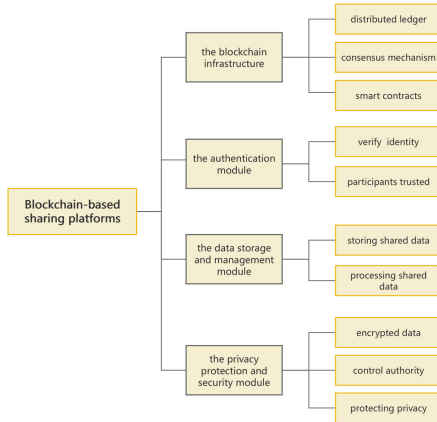
Blockchain as the Solution

- ✓ **Secure** data circulation & sharing
- ✓ **Trustworthy** decentralized consensus
- ✓ Full **traceability** of metadata
- ✓ Data “available but invisible” via privacy computing

Key insight: Blockchain immutability prevents malicious tampering or forgery of data after upload to chain.

Data Governance — Transforming stock data from chaos to order, ensuring trustworthy collection, ownership, and compliance using blockchain immutability

Component diagram of a blockchain-based sharing platform (Paper Fig. 1)



Four Platform Modules:

- **Blockchain Infrastructure** — distributed ledger, consensus, smart contracts
- **Authentication Module** — verify identity, ensure trusted participants
- **Data Storage & Management** — storing and processing shared data
- **Privacy Protection & Security** — encrypted data, access control, privacy

Real-world example: Ecological environment big data sharing using multi-chain architecture (monitoring chain, sharing chain, judicial evidence chain).

SECTION 3

Blockchain & Artificial Intelligence

Blockchain in Web3.0: Integration, Innovation & Standardization

Overview: AI in the Web3.0 Era

AI reshaping industries; blockchain enabling trustworthy AI systems

Domain	AI Application	Example
Healthcare	Medical image classification, drug discovery	PANDA: Pancreatic Cancer Detection via MRI
Transportation	Autonomous vehicles using deep RL	STRLAD: 89% completion in CARLA
Finance	High-frequency trading, credit scoring	ML pattern recognition in market data
Content (AIGC)	AI-Generated Content for Web3.0	Copyright via NFT/DAO on blockchain

AI Challenges Blockchain Solves: Data privacy & security, algorithm bias auditing, copyright of AIGC content, trustworthy communication in edge AI computing.

AIGC copyright protection and trustworthy edge AI computing

1. AIGC + Blockchain: Copyright

- AI-generated content recorded on-chain (NFT-style)
- Creators receive royalties like NFT holders
- Copyright shared in DAOs
- Eliminates infringement & misuse of assets

2. Edge AI + Blockchain: Trust

- Edge AI: decentralized, heterogeneous nodes
- Blockchain creates secure P2P environment
- Consensus avoids malicious node attacks
- Smart contracts build data marketplace

Real-world Example (Blocktrace): AI-enhanced blockchain forensics using GPT-style NL models; lets investigators query millions of transactions without data engineering skills.

AIGC — Artificial Intelligence Generated Content — AI creates text/images/code; blockchain secures copyright via immutable ledger and NFT-style DAOs

SECTION 4

Blockchain & Metaverse

Blockchain in Web3.0: Integration, Innovation & Standardization

Overview: The Metaverse and its Technology Pillars

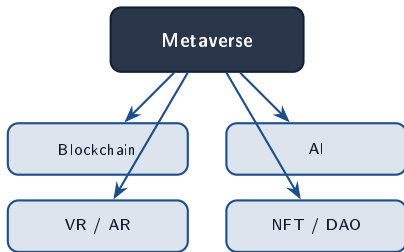
A virtual world integrating blockchain, AI, VR/AR, and digital assets

What is the Metaverse?

- Integrates VR (immersive), AR (overlay), AI (agents), Blockchain (asset management)
- Users own digital identity, virtual land, and assets
- Virtual economy: trading, work, art, social interaction

Blockchain Role in Metaverse:

- Data storage via distributed consensus
- Multi-node resource sharing
- Decentralized governance (no central authority)
- Digital currency for virtual economic transactions



“Governing the chain with the chain” — Metaverse legal framework

The Challenge:

- Metaverse contracts and transactions grow exponentially
- Need a legal framework for virtual-space activities
- Traditional courts cannot govern virtual spaces

The Solution — “Judicial Chain”:

- Alliance chain with **layered permissions** for nodes
- Data elements: property rights, governance, revenue
- Smart contracts **auto-execute** contract terms
- Real-time compliance and risk monitoring on-chain

Judicial Chain Architecture

- 1 **Ordinary Nodes** — strict judicial admission
- 2 **Senior Nodes** — management permissions, layered
- 3 **Key Nodes** — data shared by permissions
- 4 **Smart Contracts** — submissions auto-stored

Goal: Ensure safety, efficiency, and fairness in metaverse judicial processes via on-chain automation.

Proposed by: Prof. Yang Dong, Renmin Univ. (2022), National Social Science Foundation key project.

DAO (Decentralized Autonomous Organization) — A community-governed entity on blockchain where rules are encoded in smart contracts; no central management

SECTION 5

Blockchain & IoT

Blockchain in Web3.0: Integration, Innovation & Standardization

Overview: Internet of Things and its Security Challenges

Billions of connected devices demanding trust and security

What is IoT?

- Connects physical devices, sensors, objects via Internet
- Key: sensors, embedded systems, protocols (Wi-Fi, Bluetooth, LoRa), cloud/edge computing
- Applications: smart homes, agriculture, smart cities, industrial automation

IoT Security Pain Points:

- ✗ Centralized access control = single-point risk
- ✗ Communication security not guaranteed
- ✗ Malicious or compromised member nodes
- ✗ Privacy leakage of personal sensor data

Blockchain Solves IoT Trust

- **Distributed storage** eliminates single-point failure
- **Consensus mechanism** prevents malicious nodes
- **Smart contracts** automate secure data sharing
- **Incentive mechanism** encourages node cooperation
- **Immutability** ensures data integrity from source

IoT (Internet of Things) — Technology connecting physical devices, sensors, and objects via Internet, enabling real-time data collection and automated actions

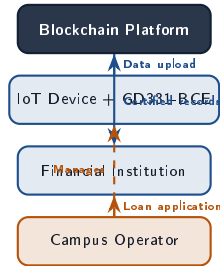
World's first trusted IoT blockchain hardware module (GD331-BCE)

Product: GD331-BCE Module

- Hardware-embedded blockchain for IoT devices
- Devices upload data to blockchain from the factory
- Capabilities: digital identity auth, data certification, tracing, privacy protection

Application: Campus BoT Finance

- IoT device operators apply for financing via platform
- On-chain data proves real device existence and capability
- Smart contracts auto-store all submission documents
- Financial institutions assess repayment from verifiable on-chain records



Impact: Reduces device maintenance costs, improves data supervision, enables rapid enterprise digital asset transformation.

SECTION 6

Blockchain & Privacy Computing

Blockchain in Web3.0: Integration, Innovation & Standardization

Overview: Privacy Computing Technologies

Three pillars of privacy computing in the data circulation era

Multi-Party Comp. (MPC)

Cryptography-Based
Collaborative compute

- Joint compute on private data
- Zero plaintext exposure
- Secret sharing & garbled circuits

Federated Learning

AI + Privacy Fusion
No raw data shared

- Train ML models locally
- Share only gradient updates
- Distributed model aggregation

Hardware TEE

Hardware Enclave
Isolated secure zone

- Hardware-level protection
- Enclave shielded from OS
- Intel SGX, ARM TrustZone

Privacy computing goal: Data is “available but invisible” — enable analysis without exposing raw personal data. Blockchain + Privacy Computing ensures compliance, immutability, and decentralization.

TEE (Trusted Execution Environment) — A hardware-isolated secure area within a processor where code executes without being readable even by privileged OS processes

Combining tamper-proof ledgers with privacy-preserving computation

The Problem:

- Massive data leaks revealed catastrophic privacy risks
- Personal data misused for fraud and criminal activities
- Digital economy requires secure, compliant data sharing

Combined Approach

- 1 Privacy computing: encrypts/de-identifies data
- 2 Blockchain: ensures tamper-proof circulation records
- 3 Combined: data flows securely and verifiably

What Blockchain Adds:

- **Tamper-proof** transaction records during data circulation
- **Traceability** — every data access logged on-chain
- **Decentralization** — no single authority controls data flow
- Smart contract-enforced **data compliance** rules
- **Incentive mechanisms** for secure data sharing

Applications: Financial fraud detection, healthcare patient data sharing (without identity exposure), market research, government sensitive statistics.

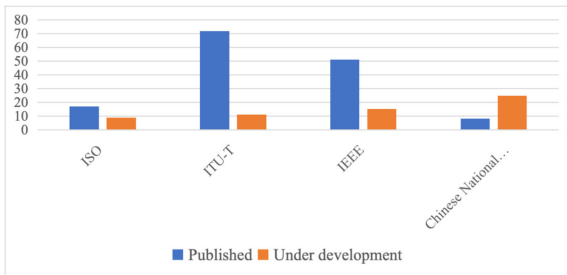
SECTION 7

Standardization Landscape

Blockchain in Web3.0: Integration, Innovation & Standardization

Overall Blockchain Standardization Status

Published and under-development standards across global bodies



Overall blockchain standardization by major organizations (paper Fig.)

Key Standardization Bodies:

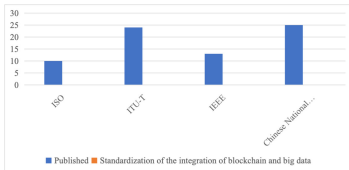
- **ITU-T** — Leading: ~72 published, ~11 under development
- **IEEE** — ~51 published, ~15 under development
- **ISO** — ~18 published, ~9 under development
- **Chinese National Standards** — ~8 published, ~25 under development

National Standard: GB/T 42752-2023 “Blockchain and Distributed Ledger Technology Reference Architecture” defines blockchain as a cryptographic-hash-chained distributed ledger.

Domain-Specific Standardization: Big Data, AI, Metaverse Standardization Landscape

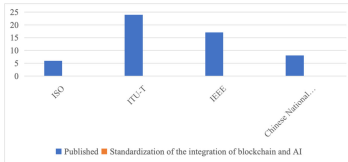
Standards progress by integration domain

Blockchain + Big Data



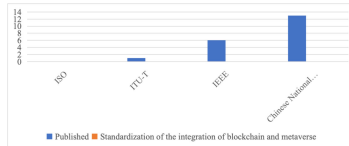
Chinese National Standards leads (~25 published). ITU-T and IEEE active.

Blockchain + AI



ITU-T leads (~24 published). Emerging domain; rapidly evolving.

Blockchain + Metaverse

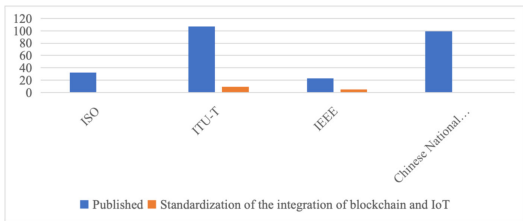


Chinese National Standards leads (~13). Youngest domain; very early stage.

Landscape

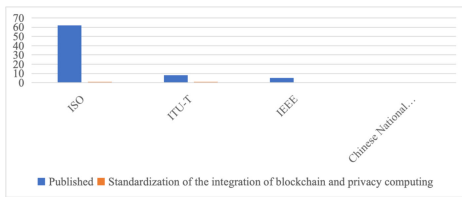
The most mature and most rapidly evolving integration standards

Blockchain + IoT Standardization



Most mature domain: ITU-T (~108) and Chinese National Standards (~100) dominate. Highest standardization activity globally.

Blockchain + Privacy Computing



ISO leads (~62 published). Reflects cryptographic heritage of privacy computing in international standards bodies.

SECTION 8

Conclusion

Blockchain in Web3.0: Integration, Innovation & Standardization

1. Foundational Role

Blockchain provides the decentralized, immutable, and traceable infrastructure on which Web3.0 is built.

3. Standardization Momentum

ITU-T, IEEE, ISO, and Chinese National Standards are actively standardizing blockchain integrations; IoT is the most mature domain.

2. Five-Domain Integration

Big Data, AI, Metaverse, IoT, and Privacy Computing are all enhanced by blockchain's trust and security guarantees.

4. Future Outlook

Blockchain will continue reshaping digital economy, privacy computing, and AIGC governance in the Web3.0 era.

SECTION 9

References

Blockchain in Web3.0: Integration, Innovation & Standardization

Key literature and foundational papers

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DOI: 10.1007/s11280-024-01319-7

[2] Bitcoin (Nakamoto)

2008

S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.

Core supporting technology of Bitcoin — blockchain

[3] GB/T 42752-2023

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"Blockchain and Distributed Ledger Technology Reference Architecture," Chinese national blockchain definition standard.

[4] PANDA — AI Healthcare

2023

Kai Cao, Yingda Xia et al., AI pancreatic cancer detection using non-contrast MRI comparable to radiology reports.

[5] GanDao GD331-BCE

2022+

World's first trusted IoT blockchain hardware module; applied in campus BoT finance, biomedicine, logistics.

[6] Yang Dong — Judicial Chain

2022

Prof. Yang Dong, Renmin University: "Governing the chain with the chain" — blockchain judicial model for the metaverse.



Thank You

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